Units 11 & 12

MARKET FAILURE AND GOVERNMENT POLICY

How can markets fail?

Conditions for markets to work well:

- **Private property =** the rights to the thing bought/sold
- Institutions e.g. government to enforce **property rights**
- **Social norms** of respecting property rights
- Ability to write complete and enforceable contracts that can be evaluated in a court of law

Markets fail when property rights are missing, incomplete, or are difficult to enforce with a contract.

Examples of Market Failure

1. Pesticides in the Caribbean
   - Banana plantation owners used harmful pesticides to reduce costs and increase their profits.
   - The chemicals leaked into the rivers and contaminated the local seafood, damaging causing residents to fall seriously ill.
2. Overuse of antibiotics
   - People often overuse antibiotics when other treatments would be better, which creates bacteria-resistant pathogens.

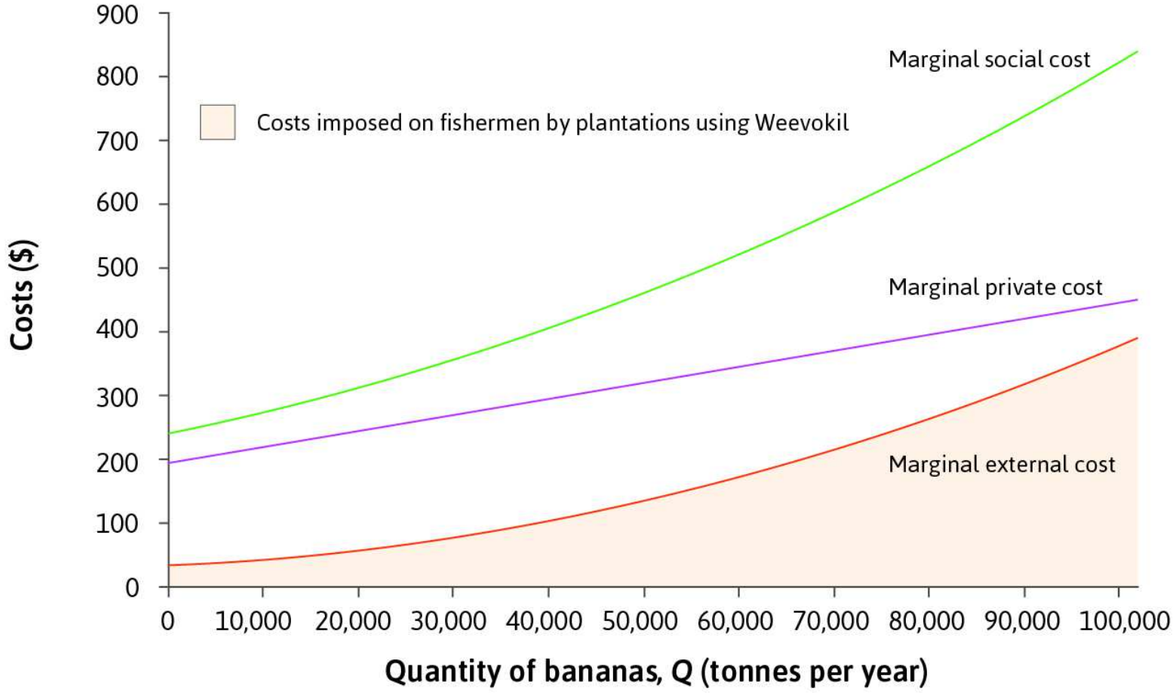
A. External Effects

Key concepts

External effect (**externality**) = an effect of an economic decision that is not specified as a benefit or liability in the contract.

Pesticide pollution example:

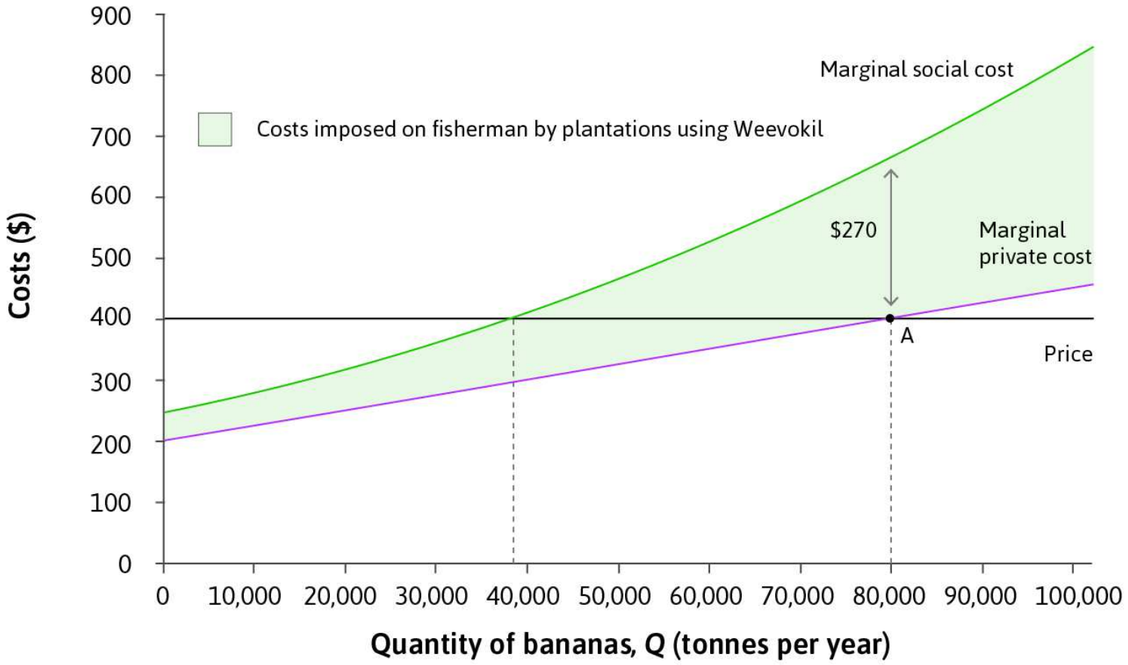
- **Marginal private cost** (MPC) = marginal cost to decision-maker
- **Marginal external cost** (MEC) = costs imposed by decision-maker on society (fishermen)
- **Marginal social cost** (MSC) = MPC + MEC (full cost to society)



negative externality (MSC > MPC)

Result: Pareto inefficiency

- Plantations produce where Price = MPC
- Pareto-efficient level is where Price = MSC
- Negative external effect leads to overproduction and overuse of pesticides.



Outcome is not Pareto efficient

At point A, fishermen could pay plantation owners up to $270 to reduce production by one unit.

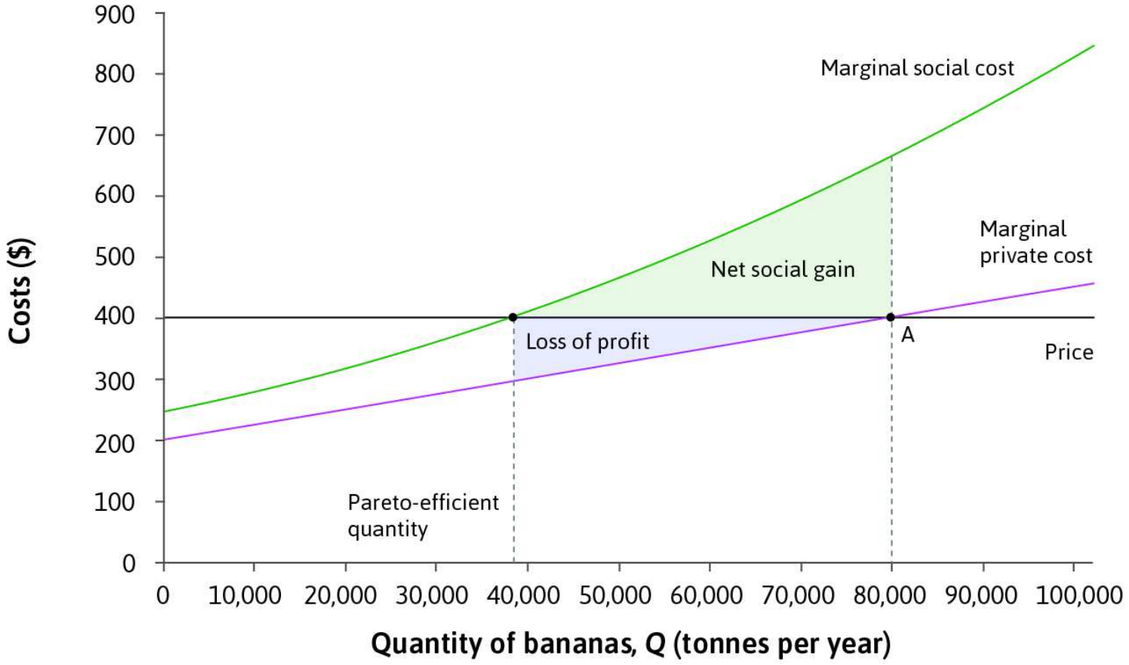
Solution #1: Bargaining

- Legally assign property rights to the externality (e.g. the right to pollute, the right to clean air)
- Private bargaining between parties involved will result in a Pareto-efficient allocation regardless of which party has the property rights, <u>in the absence of transaction costs.</u>
- May be more effective than government intervention because private parties have more of the necessary information.
- However, **transaction costs** (costs of acquiring information, enforcing the contract, or collective action) can be a major obstacle in reality.

Bargaining: Example

In the pesticides example, there is a net social gain that parties could share by reducing production, because the fall in plantations' profit is smaller than the gain for the fishermen.

- Plantation owners' **minimum acceptable offer** (minimum compensation) = lost profits.
- Fishermen's **reservation option** (maximum compensation) = the sum of green and purple areas.



Actual compensation depends on relative bargaining power.

Practical limits of bargaining

- Impediments to collective action – finding a representative and agreeing on how to split the gains within each party
- Missing information – calculating the exact costs imposed on each fisherman and each plantation's contribution to pollution.
- Enforcement – it may be difficult for a court to determine whether plantations have complied or not.
- Limited funds – fisherman may not have enough money to pay plantations the compensation required.

Solution #2: Government Policy

1) Regulation of production - cap at socially optimal amount
   - May be difficult to determine and enforce the right quota for each polluter

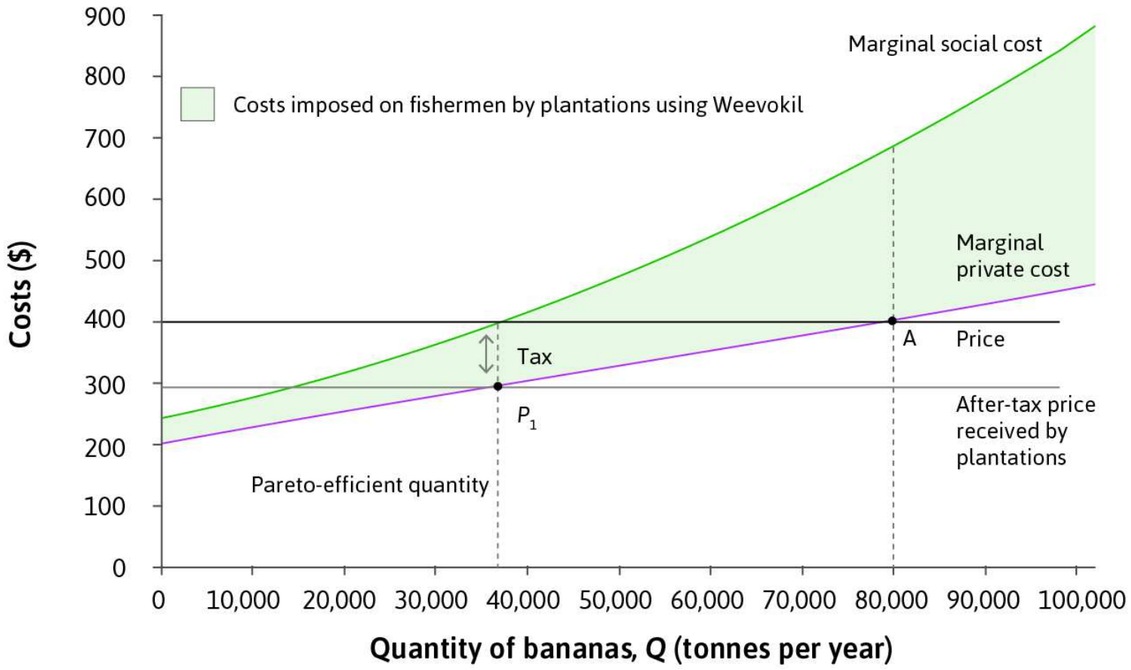
2) **Pigouvian tax/subsidy** = tax/subsidy on firms generating negative/positive external effects, in order to correct an inefficient market outcome.

3) Enforcing compensation for affected parties.

Example: Pollution tax

Government puts a per-unit tax on output, equal to the MEC.

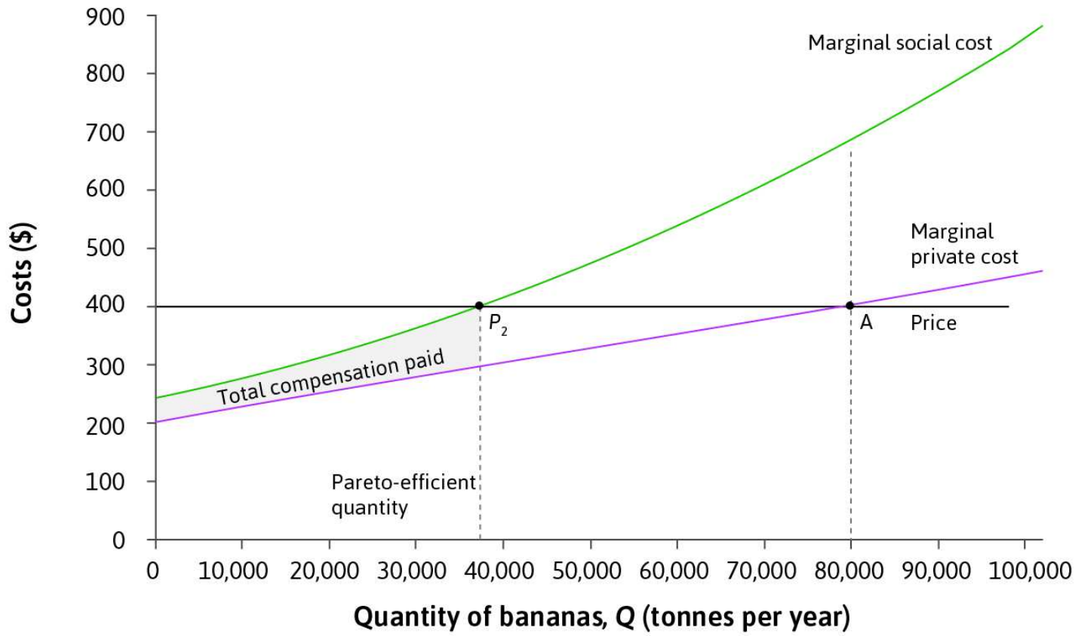
- Profit-maximizing producer chooses output where MPC = after-tax price, which is the socially optimal output.
- The tax forces producers to face the full cost of their decisions.



Example: Compensation

Government requires plantation owners to pay fishermen compensation for each tonne produced.

- Required compensation is equal to the difference between the MSC and the MPC (grey area).
- Fishermen are fully compensated, and producers choose the socially optimal level of output.



Similar effect on profits compared to a tax, but fishermen are better off (receive payment instead of the government).

Practical Limits of Policy

Similar limitations to those for private bargaining:

- Missing information – government may not be able to properly calculate marginal social cost either
- Lobbying - The government may favor the more powerful group, in which case it could impose a Pareto-efficient outcome that is unfair, or even fail to achieve efficiency

B. Other Types of Market Failure

1. Public Goods

Classification criteria: nature of good + **institutions**

Public good = **non-rival;** may or may not be **excludable**

Non-rival = use by one person does not reduce its availability to others

Non-excludable = impossible to exclude anyone from having access

	Rival	Non-rival
Excludable	Private goods (food, clothes, houses, most forms of medical care)	Public goods that are artificially scarce (subscription TV, uncongested toll roads, knowledge subject to intellectual property rights)
Non-excludable	Common-pool resources (fish stocks in a lake, common grazing land)	Non-excludable public goods and bads (view of a lunar eclipse, public broadcasts, rules of arithmetic or calculus, national defence, noise and air pollution)

Public goods and Market Failure

Markets typically allocate private goods.
But for common, public, or club goods, markets may fail:

- Non-rival goods have a marginal cost of zero, so it is not possible to set price = MC unless provision is subsidized
- It is impossible to set a price for non-excludable goods because the provider cannot exclude those who haven't paid

Example: If one farmer invests in a community irrigation project, other farmers receive a **positive externality**, but it is difficult to make them pay for the benefits or write contracts that guarantee a Pareto-efficient irrigation level

2. Asymmetric Information

When information is **asymmetric**, one party knows something relevant to the transaction but the other party does not.

Two forms of asymmetric information:

1. **Hidden action** – leads to a **moral hazard** problem

   Example – Involuntary unemployment because employers cannot observe employees' exact work effort (Unit 6).

2. **Hidden attributes** – leads to an **adverse selection** problem

   Example – Buyers of second-hand cars do not know all the attributes of the car e.g. quality, but the sellers do.

C. The Government as an Economic Actor

Markets and prices

Benefits of using markets:

- Markets allow each of us, pursuing our private objectives, to work together, producing and distributing goods and services.
- Markets allow us to interact with people in ways that we can mutually benefit from, specializing in the things we are relatively good at doing.
- When markets work well, prices send messages about the real scarcity of goods and services.

Should markets allocate everything?

Cases where markets may be undesirable:

- **Repugnant markets**: creating a market for certain goods/services would violate ethical/social norms e.g. slavery
- **Public Goods**
- When putting a price on a good or service crowds out norms of **social preferences**
- **Merit goods**: goods that may be better allocated based on talent rather than ability to pay e.g. education

Should markets allocate everything?

Cases where markets may be undesirable:

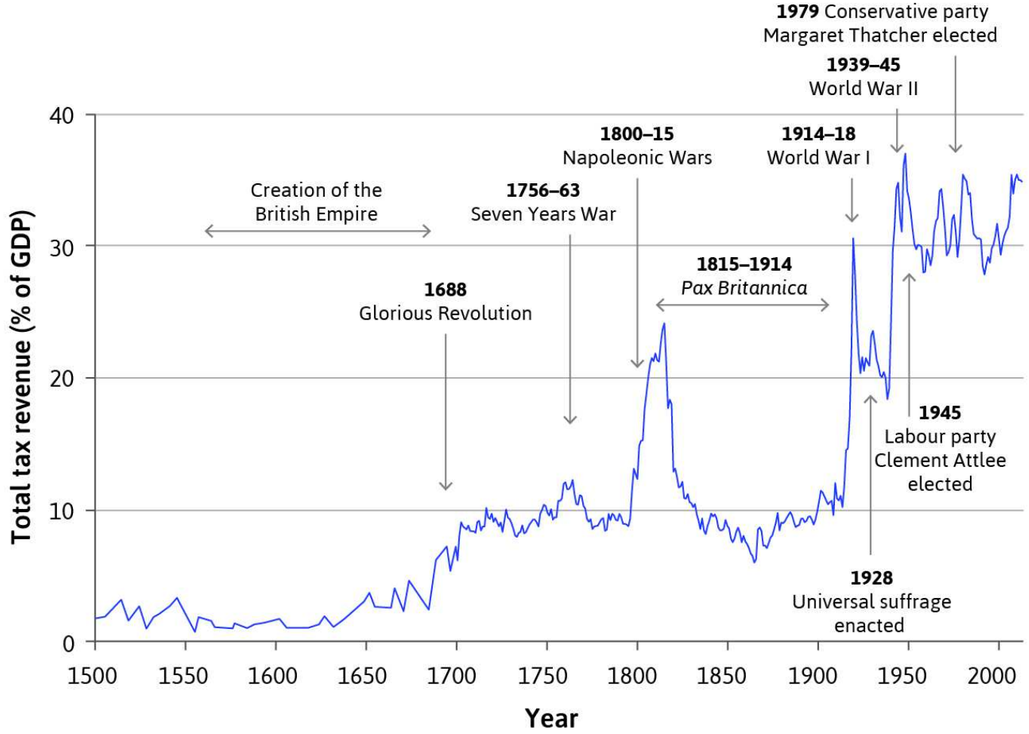
- **Repugnant markets**: creating a market for certain goods/services would violate ethical/social norms e.g. slavery
- **Public Goods**
- When putting a price on a good or service crowds out norms of **social preferences**
- **Merit goods**: goods that may be better allocated based on talent rather than ability to pay e.g. education

Government as economic actor

Government = the only entity within a territory that can dictate what people must do/not do, and can legitimately use force and restraints on an individual's freedom to achieve that end.

The government is different from other economic actors:

- Ability to levy taxes and/or issue money
- Engage in activities that can improve quality of life across a wide area



Beneficial Policy

Governments can address inefficiency and unfairness that sometimes results from private economic interactions.

Methods:

- **Incentives** – altering the costs/benefits of activities through taxes, subsidies, and other expenditures
- **Regulation** – restricting certain activities e.g. antitrust policy
- **Information** - altering available information or expectations about what others will do, to promote coordination
- **Public provision** – includes public goods and transfers

Detrimental Policy

Any organization with enough power to address issues of efficiency and fairness also has the power to magnify them.

Examples include using force to silence opponents, and rigging the rules to benefit narrow interests

Well-governed societies place limits on government power:

- Democratic elections – bad governments can be dismissed
- Constitutional restrictions on what the government can do

Governments therefore operate within constraints, like other economic actors.

D. Democracy as a Political Institution

Defining Democracy

Democracy: A form of government with 3 political features:

- Rule of law – all individuals are bound by the same laws
- Civil liberties – the rights of free speech, assembly, and the press
- Inclusive, fair, and decisive elections – fair elections in which no major population group is excluded from voting and after which the losing party leaves office.

No government perfectly satisfies these 3 criteria. But how a government actually works also depends on political preferences and electoral competition.

Dictatorship vs. Democracy

Recall the Ultimatum Game from earlier in the course

- When this game is repeated, relatively equal divisions of the money can be sustained

We also discussed a variant called the Dictator Game

- Repeating the dictator game does not result in equitable divisions since responders lack a way to punish the proposer

A functioning democracy operates like a repeated Ultimatum Game

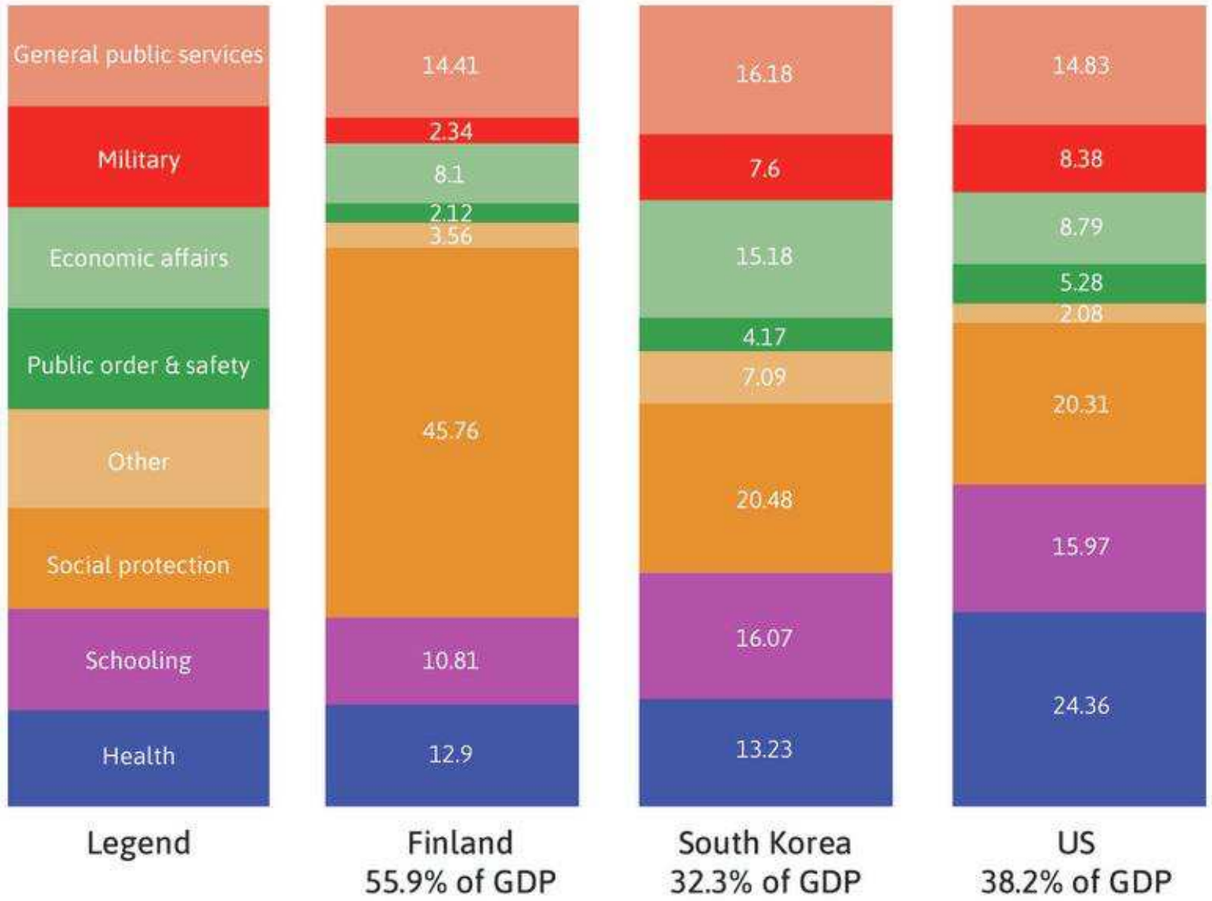
- It may be costly to displace a ruling party, but the risk of losing elections discourages corruption and self dealing

Accountability of Government

There are similarities and differences between the profit-maximizing behavior of monopoly firms
and the rent-maximizing behavior of a governing elite

Ideal Case	Constraints	Rents	Outcome
Political Monopoly	Taxing power	High	Enrichment of narrow interests
Economic Monopoly	Cost of Production	High	Price Markups
Political Competition	Electoral loss	None	Broad provision of public goods
Economic Competition	Entry of new firms	None	Profits driven to zero

Democratic spending priorities



Democratic governments differ in their spending patterns. One reason is that political institutions differ, even among democracies.

Government Failure

Government failure: A failure of political accountability

Market failures and unfairness still persist in democracies, due to:

- Economic infeasibility – there may not be a feasible set of policies that sustain a fair and efficient outcome
- Administrative infeasibility – economically feasible policies may be impossible in practice due to the limited information and capabilities of government officials
- Political infeasibility – governments may choose not to adopt feasible policies because of opposition by certain groups